

CHAPTER 03: Security in Mobile Computing

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CH03: Topics to be covered

- Overview on Security in mobile and Wireless Networks
- Security and Ad Hoc Networking Technologies
- Location Information, Security, and Privacy
- Distinguishing Privacy and Security.

- What is Security?

- What is Security?

- Protecting computer-related assets that are subject to a variety of threats



Figuring out measures to protect those assets

- The Focus of security :
 1. What assets do we need to protect?
 2. How are those assets threatened?
 3. What can we do to counter those threats?
 - According to NIST

Computer Security: Measures and controls that ensure **confidentiality**, **integrity**, and **availability** of information system assets including hardware, software, firmware, and information being processed, stored, and communicated.

- Objective of Security: → ensuring CIA

1. Confidentiality:

- Preserving authorized restrictions on information access and disclosure
- Covers two related concepts
 - **Data confidentiality** $\mapsto$ assures that private or confidential information is not made available or disclosed to unauthorized individuals.
 - **Privacy** $\mapsto$ assures that **individuals control** what information related to them may be collected and stored and by whom and to whom that information may be disclosed.

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2. Integrity

- Guarding against improper information modification or destruction
- Covers two related concepts
 - **Data integrity**: $\mapsto$ assures that information and programs are changed only in a specified and authorized manner.
 - **System integrity** $\mapsto$ Assures that a system performs its intended function in an unimpaired manner, free from deliberate or inadvertent unauthorized manipulation of the system.

3. Availability

- Assures that systems work promptly and service is not denied to authorized users

- Security is the biggest concerns when designing any application,



distributed applications $\mapsto$ "Most Mobile Apps"

- We need security for two reasons:
 1. To keep out Unauthorized users to get access to things that they are not allowed to access
 2. To ensure that computing assets access are not inadvertently given to parties not actively seeking a system breach.
- Such a breach can happen at different points:
 - Hardware
 - Software, and
 - Communication channels.
- Various types of security concerns for mobile applications $\mapsto$ OSI Model

- Security consists of four parts:
 1. Physical security
 - prevent interference signals + Protecting the hardware
 2. Data Security
 - what encryption techniques are used to protect the data.
 3. User authentication
 - how to protect the wireless network from unauthorized users.
 4. User anonymity - what kind of protection is used against information gathering

Security concerns for mobile applications

- Application Layer Security

- The most important layer for Securing our mobile application
- Think Standalone Application



Encrypting all of the data and using sufficient usernames and passwords for authentication and authorization in the application.

- Think Networked application (<https://>)



may additionally include usage of encrypted communication using techniques such as DS3

- Presentation Layer and Session Layer Security

- The most important technology $\mapsto$ SSL (Secured Socket Layer)

- Transport Layer and Network Layer Security

- These layers are, respectively, the home of TCP and IP
- The most important security tech $\mapsto$ IPSec

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- Data Link Layer Security
 - MAC (Medium Access Control) addresses belong.
- Physical Layer Security
- In addition to OSI Model



The dimensions of mobility, and user mobility leads



Other security issues that are unique from any of those concerns experienced by stationary applications:

Security concerns

- secure authentication and authorization of nodes,
- secure communications between the authenticated and authorized nodes of the network over a wireless connection
- secure deployment of an application on the target device
- securing any conversion of content required for supporting multimodal applications
- defending against the fraudulent usage of the wireless service, and
- defending against various Denial of Service attacks that may interrupt service to the network users

CH03: Wirelss Security

- Recall Wireless Network !
- Think the advantage of such network !
- Categorized in to three
 - Wireless Wide Area Networks (WWAN)
 - Wireless Metropolian Area Networks (WMAN),
 - Wireless Local Area Networks,(WLAN) and
 - Wireless Personal Area Networks (WPAN).
- Questions $\mapsto$ what is the key problem with wireless communication ?

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Via LAYERED SECURITY Approach

CH03: Wirelss Security

- A layered approach to wireless security can provide a high degree of protection
- The layered approach has four levels:
 1. **Level - 1** $\mapsto$ **Wireless deployment and policy**
 - Best Practice $\uparrow$
 - Deploy the minimum number of **WAPs** needed for adequate Coverage.
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 - Installation of WAPs
 - NIC operational mode $\mapsto$ **infrastructure mode** and **adhoc mode**
 - WLAN user-group access, including employees, visitors
 2. **Level - 2** $\mapsto$ **Wireless access control**
 - **You Know what Access Control!**
 - Access control has two components:
 - **Device access control** and **user authentication (personnel access control)**
 - Level-2 is concerned with **device access**

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By properly configuring and managing WEP and SSID

- Best practices :
 - Configure the WEP for the highest level of encryption.
 - Change the SSID regularly, where practical.
 - Do not broadcast the SSID.
 - Verify the media access control (MAC) address upon device connection.

3 Level - 3 $\mapsto$ Perimeter security

- is protecting the wireless network Infrastructure.
- **Virtual private Network (VPN)**



is an extension of a private network that encompasses **encapsulated**, **encrypted** and **authenticated** connections.

- **VPN technology** $\mapsto$ provides a method for securing traffic
- Best practices
 - Install an **intrusion prevention system (IPS)** and wireless firewall on WLAN.
 - Encrypt **WLAN traffic using a virtual private network (VPN)**.
 - Direct all traffic through the **VPN server** and configure clients appropriately.
 - Maintain and enforce **VPN routing** and **access policies**.
 - **Maintain** and **enforce access policies for user authentication** (i.e, username/password)

4 Level - 4 $\mapsto$ Application security

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Key Protection $\mapsto$ How ? $\mapsto$ Secure Software development life-cycle

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Key Protection $\mapsto$ How ? $\mapsto$ Secure Software development life-cycle



- Level - 4 $\mapsto$ Application security ...
 - Best practices for application security include:
 - Implement an application-level user authentication system.
 - Maintain and enforce permissions and password policies.
 - Install vendor patches as they become available

$\Rightarrow$ WIRELESS NETWORK COMPONENTS AND ARCHITECTURAL MODELS

- Wireless Network architecture has two components:
 1. Wireless end station
 - Any physical component (device) that can communicate using the 802.11 standard.
 2. Access points
 - A physical component (device) that can provide two functions:
 - It acts as a network platform for connections between WLANs or to a wired LAN and
 - As a relay between stations attached to the same AP.

→ The IEEE 802.11 standard also defines 2 design structures

- **Infrastructured Mode** ↪ "Relay on Fixed Infrastructures such as AP".
- **Ad-Hoc Mode**
 - decentralized type of network. ↪ Means
↓
"Theoretically they can be formed any time and anywhere, allowing multiple users to create wireless connections cheaply, quickly, and easily with minimal hardware and user maintenance"
- Securing wireless ad hoc networks is a cumbersome task ↪ Because
 - **Vulnerability of channels**
 - message can be eavesdropped and fake messages can be injected into the network
 - **Vulnerability of nodes**
 - **Absence of infrastructure**
 - **Dynamically changing topology**

⊨ Results with threats and attacks such as **Masquerade**, **Modification of messages**, **Denial of service**, **Eavesdropping** and other

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such as **authentication**, **confidentiality**, **integrity** and availability to mobile users. ↳ **How to achieve ?**



By implementing security mechanisms that provide complete protection spanning the entire protocol stack

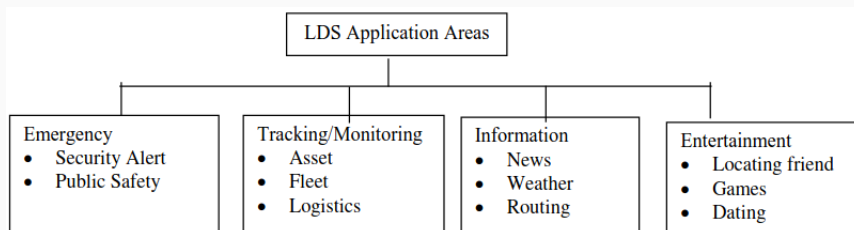
- What are security issues in protocol suite?
 - **Application Layer** : detecting and preventing viruses, worms, malicious codes and application abuses
 - **Transport layer**: Authenticating and securing end-to-end communications through data encryption
 - **Network layer**: Protecting the ad hoc routing and forwarding protocols
 - **Link layer** : protecting the wireless MAC protocol and providing link-layer security support
 - **Physical layer**: Preventing signal jamming denial-of-service attacks

- You know w@ Location Information is ?
 - are key in Ubiquitous computing $\mapsto$ **Anything, Anytime at Anywhere**
 - As a result of **LI** and **advances** in Mobile Tech
 - $\downarrow$
 - new set of service application invented $\mapsto$ LDS

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- biggest hurdles in making location information available to mobile applications



security and privacy concerns

- most of the security & privacy intimidations related to LDS have still not been appropriately addressed
- What are these security Concerns ?
 - Authentication:
 - Only authenticated users are supposed to grant permission when they request for joining the LDS
i.e. when the Service Provider (SP) is requested for location information by the user through a connecting station.
 - Traceability:
 - Any user who mishandles the network is supposed to get traceable and reported by an authority.
 - This helps in reducing the impact of the misbehaving user all across the network
 - Confidentiality:
 - It assures that the message delivery is done to authorized parties only.
 - It can be achieved through a secure key management system's encryption techniques.
 - Efficiency:
 - A lightweight management scheme should be used to ensure effective operation.

- What about the Privacy Concerns ?

- Anonymity:

- The ability of a user to access something without disclosing the identity to the third-parties.

- Short-term linkability:

- A receiving vehicle (VANET) must be capable of verifying that the multiple messages it is receiving within a small-time frame.

- Pseudonymity

- It gives assurance of resource access by a user without sharing its identity, but can still be answerable for that use.

- Location Privacy:

- The location of the user must be shielded from unapproved access.

CH03: Distinguishing Privacy and Security.

- Security and Privacy
 - **Privacy** is the right to control how your information is viewed and used
 - **Security** is protection against threats or danger.
- developer faces two distinct challenges:
 - **delivering a secure application** and
 - **delivering a private application.**
- Privacy means very different things in different countries and is subject to country-specific sets of laws
- Privacy within the realm of **wirelessly connected systems** to be of four components
 - **Data Privacy**: The contents of the transaction should be protected from disclosure to an unauthorized party.
 - **Source and Destination Privacy**: The parties involved in the transaction should not be revealed to an unauthorized party.
 - **Location Privacy**: The location of the parties making the transaction, whether physical (geographical) or logical (with respect to the network), should not be disclosed to an unauthorized party.
 - **Time Privacy**: The exact time when a transaction occurs should not be disclosed to an unauthorized party

- Tips for protecting your privacy & security
 - Limit your social media presence and the data you share online.
 - Read the organization's privacy policy in its entirety before agreeing and accepting the terms.
 - Keep your social security number (SSN) secure, and avoid distributing it when possible.
 - Use multifactor authentication to log in to secure sites. Install and use security software.
 - Use a VPN when on public WiFi.
 - Ensure your router is secure, and use a firewall.
 - Consider identity theft protection services.
 - Use different passwords for different sites and applications. Make each password complex, or consider a passphrase instead.

Done!
Question!